

Introduction to machine learning 89-511,
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Home Assignment 1

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1 Correlation and independence

- A. Let X and Y be two independent random variables. Prove that they are also uncorrelated, that is, prove that $E[XY] = E[X]E[Y]$, or $Cov(X, Y) = 0$.

Solution: By definition, two random variables X and Y are independent iff their joint probability distribution can be expressed as the product of their marginal distributions. We will show that independence implies $E[XY] = E[X]E[Y]$ for both discrete and continuous cases.

Discrete case: For discrete random variables, the expected value of XY is:

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy \cdot P(X = x, Y = y) \\ &= \sum_x \sum_y xy \cdot P(X = x)P(Y = y) \quad (\text{by independence}) \\ &= \sum_x x \cdot P(X = x) \sum_y y \cdot P(Y = y) \\ &= E[X] \cdot E[Y] \end{aligned}$$

Continuous case: For continuous random variables with joint density function $f_{X,Y}(x, y)$ and marginal densities $f_X(x)$ and $f_Y(y)$:

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f_X(x)f_Y(y) dx dy \quad (\text{by independence}) \\ &= \int_{-\infty}^{\infty} yf_Y(y) \left(\int_{-\infty}^{\infty} xf_X(x) dx \right) dy \quad (\text{treating } yf_Y(y) \text{ as constant w.r.t. } x) \\ &= \int_{-\infty}^{\infty} xf_X(x) dx \int_{-\infty}^{\infty} yf_Y(y) dy \quad (\text{treating } \int_{-\infty}^{\infty} xf_X(x) dx \text{ as a constant w.r.t. } y) \\ &= E[X] \cdot E[Y] \end{aligned}$$

Since $Cov(X, Y) = E[XY] - E[X]E[Y]$, we have $Cov(X, Y) = 0$, which means X and Y are uncorrelated.

- B. Give an example of two uncorrelated random variables that are not independent.

Solution: Let X be a random variable that takes values in $\{-1, 0, 1\}$ with equal probability $P(X = -1) = P(X = 0) = P(X = 1) = \frac{1}{3}$. Define $Y = X^2$.

First, let's calculate the expected values:

$$\begin{aligned} E[X] &= \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0 \\ E[Y] &= E[X^2] = \frac{1}{3}(-1)^2 + \frac{1}{3}(0)^2 + \frac{1}{3}(1)^2 = \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{2}{3} \\ E[XY] &= E[X \cdot X^2] = E[X^3] = \frac{1}{3}(-1)^3 + \frac{1}{3}(0)^3 + \frac{1}{3}(1)^3 = \frac{1}{3}(-1) + 0 + \frac{1}{3}(1) = 0 \end{aligned}$$

Therefore, $\mathbb{E}[XY] = 0 = \mathbb{E}[X] \cdot \mathbb{E}[Y] = 0 \cdot \frac{2}{3} = 0$, which means X and Y are uncorrelated.

However, X and Y are clearly not independent because knowing the value of X completely determines the value of Y . For instance, if we know $X = 0$, then Y must equal 0 as well. More formally, $P(X = 0, Y = 0) = P(X = 0) = \frac{1}{3} \neq P(X = 0) \cdot P(Y = 0) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$. This shows that X and Y are not independent.

- C. Give an example of three random variables such that every pair of variables is independent, but any variable can be completely determined if the two other variables are known.

Solution: Let X and Y be independent Bernoulli random variables with $P(X = 0) = P(X = 1) = \frac{1}{2}$ and $P(Y = 0) = P(Y = 1) = \frac{1}{2}$. Define $Z = X \oplus Y$ (XOR operation, i.e., $Z = X + Y \pmod{2}$).

Pairwise independence:

- X and Y are independent by definition.
- For X and Z :

$$\begin{aligned} P(X = 0, Z = 0) &= P(X = 0, X \oplus Y = 0) \\ &= P(X = 0, Y = 0) \\ &= P(X = 0)P(Y = 0) \quad (\text{since } X \text{ and } Y \text{ are independent}) \\ &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \end{aligned}$$

Also,

$$\begin{aligned} P(X = 0)P(Z = 0) &= P(X = 0)P(X \oplus Y = 0) \\ &= P(X = 0)[P(X = 0, Y = 0) + P(X = 1, Y = 1)] \\ &= P(X = 0)[P(X = 0)P(Y = 0) + P(X = 1)P(Y = 1)] \\ &= \frac{1}{2} \left[\frac{1}{4} + \frac{1}{4} \right] = \frac{1}{4} \end{aligned}$$

So $P(X = 0, Z = 0) = P(X = 0)P(Z = 0)$. Similarly, we can show that pairs (X, Z) and (Y, Z) are independent.

Determinism from two variables: Z is determined by X and Y ($Z = X \oplus Y$). Knowing X and Z determines Y because $Y = X \oplus Z$. Knowing Y and Z determines X because $X = Y \oplus Z$. So, X, Y, Z are pairwise independent, but any variable is determined by the other two.

- D. Let $x_1, \dots, x_n$ be a set of column vectors of dimension $d > n$. Let C be an (empirical) population correlation matrix $C = \sum_{i=1}^n x_i x_i^\top$. Note that this is not a covariance matrix unless the means are zero.

Prove that the matrix C is positive semi-definite. That is, for any vector $z \in \mathbb{R}^d$, it holds that $z^\top C z \geq 0$.

Solution:

$$z^\top C z \stackrel{(1)}{=} z^\top \left(\sum_{i=1}^n x_i x_i^\top \right) z \stackrel{(2)}{=} \sum_{i=1}^n z^\top x_i x_i^\top z \stackrel{(3)}{=} \sum_{i=1}^n (z^\top x_i) (x_i^\top z) \stackrel{(4)}{=} \sum_{i=1}^n (z^\top x_i) (z^\top x_i)^\top \stackrel{(5)}{=} \sum_{i=1}^n (z^\top x_i)^2 \stackrel{(6)}{\geq} 0$$

(1): Definition of $C = \sum_{i=1}^n x_i x_i^\top$

(2): Distributive property.

(3): Associativity of matrix multiplication.

(4): Since $x_i^\top z = (z^\top x_i)^\top$

(5): For a scalar a , $a^\top = a$, and $z^\top x_i$ is a scalar.

(6): Sum of non-negative squares.

2 Multivariate Gaussians

Let $X \sim N(\mu, \Sigma)$ be a standard multivariate Gaussian of dimension d . That is, its mean is the vector of zeros $\mu = 0$ and its covariance matrix is the d -dimensional identity matrix $\Sigma = I_d$.

- A. For $d = 1$, what is the probability that a sample is within one standard deviation from the mean (zero)?

Solution: For $d = 1$, we have $X \sim N(0, 1)$. Therefore, the probability that a sample is within one standard deviation from the mean is given by:

$$P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 \approx 0.6827$$

- B. Prove that as $d \rightarrow \infty$ the probability that a sample is within a unit distance from the center decays to zero. Hint, the d -dimensional unit sphere around zero is contained within a d -dimensional cube around zero with a edge side of 2. Use it to bound the probability that a sample is inside the cube in every dimension.

Solution: We want to show that $\lim_{d \rightarrow \infty} \mathbb{P}(\|X\| \leq 1) = 0$, where $X \sim N(\mathbf{0}, I_d)$.

Let $B_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \leq 1\}$ be the d -dimensional unit ball and $C_d = \{\mathbf{x} \in \mathbb{R}^d : |x_i| \leq 1 \text{ for all } i = 1, \dots, d\}$ be the d -dimensional (hyper)cube of side length 2, both centered at the origin.

First, we will show that $B_d \subseteq C_d$. Suppose $\mathbf{x} \in B_d$. By definition, $\|\mathbf{x}\|^2 = \sum_{j=1}^d x_j^2 \leq 1$. Since each term x_j^2 in the sum is non-negative ($x_j^2 \geq 0$), it follows that for any specific component i , we must have $x_i^2 \leq \sum_{j=1}^d x_j^2$. Putting it together, we get $x_i^2 \leq \sum_{j=1}^d x_j^2 \leq 1$. Taking the square root of both sides gives $\sqrt{x_i^2} = |x_i| \leq \sqrt{1} = 1$. Since this holds for all components $i = 1, \dots, d$, the condition $|x_i| \leq 1$ for all i is satisfied. This is exactly the definition of C_d , so $\mathbf{x} \in C_d$. Therefore, we have shown $B_d \subseteq C_d$.

This containment lets us bound probability that a sample is within a unit distance from the center by the probability that it is inside the cube C_d :

$$\mathbb{P}(\|X\| \leq 1) = \mathbb{P}(X \in B_d) \leq \mathbb{P}(X \in C_d).$$

The event $X \in C_d$ happens if and only if $|X_i| \leq 1$ for all components $i = 1, \dots, d$. Since $X \sim N(\mathbf{0}, I_d)$, its covariance matrix is the identity matrix I_d . As established in the lecture, a diagonal covariance matrix for a multivariate normal distribution implies that the components $X_1, \dots, X_d$ are independent Gaussian variables. Furthermore, since the mean is $\mathbf{0}$ and the variances are all 1, each component follows a standard normal distribution: $X_i \sim N(0, 1)$ for all i .

Because the components X_i are independent, we can calculate the probability of the joint event as the product of the individual probabilities:

$$\mathbb{P}(X \in C_d) = \mathbb{P}(|X_1| \leq 1, \dots, |X_d| \leq 1) = \prod_{i=1}^d \mathbb{P}(|X_i| \leq 1).$$

Let $p = \mathbb{P}(|X_i| \leq 1)$. Since each $X_i \sim N(0, 1)$, this probability p is the same for all i . From part A, we know $p = \Phi(1) - \Phi(-1) \approx 0.6827$. Note that p is a constant satisfying $0 < p < 1$.

Substituting this probability back, we get:

$$\mathbb{P}(X \in C_d) = p^d.$$

Combining this result with our earlier probability inequality:

$$0 \leq \mathbb{P}(\|X\| \leq 1) \leq p^d.$$

Since $0 < p < 1$, the limit of p^d as d approaches infinity is 0: $\lim_{d \rightarrow \infty} p^d = 0$. By the Squeeze Theorem, since $\mathbb{P}(\|X\| \leq 1)$ is non-negative and bounded above by a quantity (p^d) that goes to 0, the probability itself must also converge to 0.

Therefore, we conclude:

$$\lim_{d \rightarrow \infty} \mathbb{P}(\|X\| \leq 1) = 0.$$

3 Linear transformation of the covariance matrix

Let X be a random vector $X = [X_1, \dots, X_d]$, with a population covariance matrix C . Let w be some vector in $\mathbb{R}^d$ and define $y = w^\top X$. Show that the variance of y is $\text{Var}(y) = w^\top C w$.

Solution:

Let $\mu = E[X] = (\mu_1, \dots, \mu_d)^\top$. The covariance matrix C is defined as $C = E[(X - \mu)(X - \mu)^\top]$. The (i, j) -th element of C is $C_{ij} = \text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$.

First, let's find the expected value of y :

$$E[y] = E[w^\top X] = E\left[\sum_{i=1}^d w_i X_i\right] = \sum_{i=1}^d w_i E[X_i] = w^\top E[X] = w^\top \mu$$

Now, we can calculate the variance of y using the definition $\text{Var}(y) = E[(y - E[y])^2]$.

$$\text{Var}(y) = E[(y - E[y])^2] = E[(w^\top X - w^\top \mu)^2] = E[(w^\top (X - \mu))^2]$$

Since $w^\top (X - \mu)$ is a scalar, its square is $(w^\top (X - \mu))(w^\top (X - \mu))$. We can also write the second term as $((X - \mu)^\top w)$, because the transpose of a scalar is itself.

$$\text{Var}(y) = E[(w^\top (X - \mu))((X - \mu)^\top w)]$$

Since w is a constant vector, we can move it outside the expectation in a specific way. Using linearity of expectation and the property $E[aZb] = aE[Z]b$ for constant vectors a, b :

$$\text{Var}(y) = w^\top E[(X - \mu)(X - \mu)^\top] w$$

Recognizing the definition of the covariance matrix $C = E[(X - \mu)(X - \mu)^\top]$:

$$\text{Var}(y) = w^\top C w$$

4 Eigen decomposition

Let A be an $n \times n$ square matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.

- a. Prove that A and its transpose $A^\top$ have the same set of eigenvalues.

Solution: The eigenvalues are the roots of the characteristic polynomial. The characteristic polynomial of A is $p_A(\lambda) = \det(A - \lambda I)$. The characteristic polynomial of $A^\top$ is $p_{A^\top}(\lambda) = \det(A^\top - \lambda I)$. We use the property that $\det(M) = \det(M^\top)$ for any square matrix M . Let $M = A - \lambda I$. Then $M^\top = (A - \lambda I)^\top = A^\top - (\lambda I)^\top = A^\top - \lambda I$. Therefore,

$$p_A(\lambda) = \det(A - \lambda I) = \det((A - \lambda I)^\top) = \det(A^\top - \lambda I) = p_{A^\top}(\lambda)$$

Since A and $A^\top$ have the same characteristic polynomial, they have the same roots, which are the eigenvalues. Therefore, A and $A^\top$ have the same set of eigenvalues.

- b. Assume that A is diagonalizable. Prove that the determinant of A is given by the product of its eigenvalues: $\det(A) = \prod_{i=1}^n \lambda_i$.

Solution: Since A is diagonalizable, there exists an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. The diagonal entries of D are the eigenvalues of A , so $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. Using the properties $\det(XY) = \det(X)\det(Y)$ and $\det(P^{-1}) = 1/\det(P)$, we compute the determinant of A :

$$\det(A) = \det(PDP^{-1}) = \det(P)\det(D)\det(P^{-1}) = \det(P)\det(D)\frac{1}{\det(P)} = \det(D)$$

The determinant of a diagonal matrix is the product of its diagonal entries:

$$\det(D) = \lambda_1 \lambda_2 \cdots \lambda_n = \prod_{i=1}^n \lambda_i$$

Therefore, $\det(A) = \prod_{i=1}^n \lambda_i$.

5 Gradients

Below are two functions of the form $f(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ that take an input vector $x \in \mathbb{R}^d$. State the range where the function is differentiable. Compute their gradients with respect to x . In both cases, $w \in \mathbb{R}^d$ is a given vector of constants.

a.

$$f(w, x) = \begin{cases} 0 & \text{if } w^\top x \geq 1, \\ 1 - w^\top x & \text{otherwise} \end{cases}$$

Solution:

We consider two cases for the constant vector w .

Case 1: $w = 0$ (the zero vector). If $w = 0$, then $w^\top x = 0$ for all x . Since $0 < 1$, the function becomes $f(0, x) = 1 - 0 = 1$ for all x . This is a constant function.

- **Differentiability:** Differentiable everywhere (for all $x \in \mathbb{R}^d$).
- **Gradient:** The gradient of a constant function is the zero vector: $\nabla_x f(0, x) = 0$.

Case 2: $w \neq 0$. We examine the function in different regions based on the value of $w^\top x$.

- **Region** $w^\top x > 1$: Here, $f(w, x) = 0$. This is a constant value. The function is differentiable in this region, and its gradient is $\nabla_x f(w, x) = 0$.
- **Region** $w^\top x < 1$: Here, $f(w, x) = 1 - w^\top x$. Its gradient is found by taking partial derivatives: $\frac{\partial}{\partial x_j}(1 - \sum_{i=1}^d w_i x_i) = -w_j$. Therefore, the gradient is $\nabla_x f(w, x) = -w$ and the function is differentiable in this region.
- **Boundary** $w^\top x = 1$: To check differentiability here, we compare the gradients as we approach the boundary from each region. From the region $w^\top x > 1$, the gradient is 0. From the region $w^\top x < 1$, the gradient is $-w$. Since $w \neq 0$, the gradient 0 and the gradient $-w$ are different. Because the gradient is different depending on which side you approach from, the function is not differentiable at any point x where $w^\top x = 1$.

Summary:

- If $w = 0$, the function is differentiable for all $x \in \mathbb{R}^d$.
- If $w \neq 0$, the function is differentiable for all x such that $w^\top x \neq 1$.

The gradient, where it exists, is:

$$\nabla_x f(w, x) = \begin{cases} 0, & \text{if } w^\top x > 1, \\ -w, & \text{if } w^\top x < 1. \end{cases}$$

(Note: If $w = 0$, this formula gives $\nabla_x f = -0 = 0$ for $w^\top x = 0 < 1$, consistent with the $w = 0$ case.)

b.

$$f(x) = \frac{1}{1 + e^{-w^\top x}}.$$

Show that the gradient w.r.t. x equals $\nabla f(x) = f(x)(1 - f(x))w$.

Solution:

To find the gradient $\nabla_x f(x)$ we will use the chain rule:

$$\begin{aligned} \nabla_x f(x) &= \nabla_x \left(\frac{1}{1 + e^{-w^\top x}} \right) \\ &= -(1 + e^{-w^\top x})^{-2} \cdot \nabla_x (1 + e^{-w^\top x}) \\ &= -(1 + e^{-w^\top x})^{-2} \cdot e^{-w^\top x} \cdot \nabla_x (-w^\top x) \\ &= -(1 + e^{-w^\top x})^{-2} \cdot e^{-w^\top x} \cdot (-w) \\ &= \frac{e^{-w^\top x}}{(1 + e^{-w^\top x})^2} w \end{aligned}$$

To show this equals the desired form, we can rewrite it as:

$$\nabla_x f(x) = \left(\frac{1}{1 + e^{-w^\top x}} \right) \left(\frac{e^{-w^\top x}}{1 + e^{-w^\top x}} \right) w$$

Notice that $f(x) = \frac{1}{1 + e^{-w^\top x}}$.

Also, $1 - f(x) = 1 - \frac{1}{1 + e^{-w^\top x}} = \frac{(1 + e^{-w^\top x}) - 1}{1 + e^{-w^\top x}} = \frac{e^{-w^\top x}}{1 + e^{-w^\top x}}$.

Substituting these into the expression for $\nabla_x f(x)$, we get:

$$\nabla_x f(x) = f(x)(1 - f(x))w.$$

The function is differentiable for all x